Lambek Calculus and Formal Languages

(Extended abstract)

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Introduction

The systematic study of generating grammars was started by N. Chomsky in the 50s (cf. [10]). He defined several classes of generating grammars, which are interesting for both linguists and mathematicians, e. g. context-sensitive grammars, context-free grammars, and linear grammars. On the other hand, categorial grammars were studied by Y. Bar-Hillel, J. Lambek and others. The notion of a basic categorial grammar was introduced in [1]. In the same paper it was proved that the languages recognized by basic categorial grammars are precisely the context-free ones.

Another kind of categorial grammar was introduced by J. Lambek [15]. These grammars are based on a syntactic calculus, presently known as the Lambek calculus. Chomsky [11] conjectured that these grammars are also equivalent to context-free ones. In [12] Cohen proved that every basic categorial grammar (and, thus, every context-free grammar) is equivalent to a Lambek grammar. He also proposed a proof of the converse. However, as pointed out in [6], this proof contains an error. Buszkowski proved that some special kinds of Lambek grammars are context-free [6, 8, 9]. These grammars use weakly unidirectional types or types of order at most two.

The first result of this paper (Theorem 1) says that Lambek grammars generate only context-free languages. Thus they are equivalent to context-free grammars and also to basic categorial grammars. This fact (sometimes called *the Chomsky Conjecture*) was proved in [16] and [17].

The inteneded syntactic string models, i.e., *free semigroup models* (also called *language models* or *L-models*) for the Lambek calculus were considered in [3], [4], and [5]. The more general class of *groupoid models* has been studied in [7], [13], and [14]. In [4] W. Buszkowski established that the product-free fragment of the Lambek calculus is L-complete (i.e., complete w.r.t. free semigroup models), using the canonical model. The question of L-completeness of the full Lambek calculus remained open (cf. [2]).

The second result of this paper (Theorem 2) gives a positive answer to this question. The proof has been publised in [18] and [19].

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1 Preliminaries

For any set $\mathcal{M}$ we denote by $\mathcal{M}^+$ the set of all finite non-empty strings consisting of elements of $\mathcal{M}$. The set of all subsets of $\mathcal{M}$ is denoted by $\mathbf{P}(\mathcal{M})$.

We consider the syntactic calculus introduced in [15]. The types of the Lambek calculus are built of primitive types $p_1, p_2, \ldots$ and three binary connectives $\bullet, \backslash, /$. We shall denote the set of all types by Tp. Capital letters $A, B, \ldots$ range over types. Capital Greek letters range over finite (possibly empty) sequences of types. Sequents of the Lambek calculus are of the form $\Gamma \to A$, where Γ is a nonempty sequence of types.

Axioms: $A \to A$

Rules:

$$\frac{\Gamma \to A \quad \Delta \to B}{\Gamma\Delta \to A \bullet B}\ (\to\bullet) \qquad \frac{\Gamma A B \Delta \to C}{\Gamma (A \bullet B) \Delta \to C}\ (\bullet\to)$$

$$\frac{A\Pi \to B}{\Pi \to A \backslash B}\ (\to\backslash) \text{ where } \Pi \text{ is non-empty} \qquad \frac{\Pi \to A \quad \Gamma B \Delta \to C}{\Gamma \Pi (A \backslash B) \Delta \to C}\ (\backslash\to)$$

$$\frac{\Pi A \to B}{\Pi \to B / A}\ (\to/) \text{ where } \Pi \text{ is non-empty} \qquad \frac{\Pi \to A \quad \Gamma B \Delta \to C}{\Gamma (B/A) \Pi \Delta \to C}\ (/\to)$$

$$\frac{\Pi \to B \quad \Gamma B \Delta \to A}{\Gamma \Pi \Delta \to A}\ (CUT)$$

The cut-elimination theorem for this calculus is proved in [15].

2 Lambek grammars recognize context-free languages

Definition. We assume that a finite alphabet $\mathcal{T}$ and a distinguished type D are given. A *Lambek grammar* is a mapping f such that, for all $t \in \mathcal{T}$, $f(t) \subset \mathrm{Tp}$ and $f(t)$ is finite.

The *language generated by the Lambek grammar* is defined as the set of all expressions $t_1 \ldots t_n$ over the alphabet $\mathcal{T}$ for which there exists a derivable sequent $B_1 \ \ldots \ B_n \to D$ such that $B_i \in f(t_i)$ for all $i \leq n$.

Definition. We assume that two disjoint alphabets $\mathcal{T}$ and $\mathcal{W}$ are given. The elements of $\mathcal{T}$ are called *terminal symbols* and those of $\mathcal{W}$ are *auxiliary symbols.*

A *context-free rewrite rule* is of the form $X \Rightarrow e$, where X is an auxiliary symbol and e is a non-empty word in the alphabet $\mathcal{T} \cup \mathcal{W}$.

A *context-free grammar* is a finite set $\mathcal{R}$ of context-free rewrite rules, with one auxiliary symbol S designated as its *start symbol.*

By $\bar{\mathcal{G}}(\mathcal{T}, \mathcal{W}, S, \mathcal{R})$ we denote the set of all expressions over the alphabet $\mathcal{T} \cup \mathcal{W}$ that arise through some finite sequence of rewritings of the start symbol S via the rules of $\mathcal{R}$.

The *language generated by the context-free grammar* is defined as

$$\bar{\mathcal{G}}(\mathcal{T}, \mathcal{W}, S, \mathcal{R}) \cap \mathcal{T}^+.$$

Theorem 1. *For any Lambek grammar there exists a context-free grammar such that the languages generated by these grammars coincide.*

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3 L-completeness of the Lambek calculus

Definition. We define *L-model* (also called *language model* or *free semigroup model*) to be a triplet $\langle \mathcal{W}^+, \circ, w \rangle$, where $\mathcal{W}$ is an arbitrary alphabet, $\circ$ denotes concatenation of words from $\mathcal{W}^+$, and w is a function $w\colon \mathrm{Tp} \to \mathbf{P}(\mathcal{W}^+)$ such that

(1) $w(A \bullet B) = w(A) \circ w(B)$;
(2) $w(A \backslash B) = \{\gamma \in \mathcal{W}^+ \mid w(A) \circ \{\gamma\} \subseteq w(B)\}$;
(3) $w(B/A) = \{\gamma \in \mathcal{W}^+ \mid \{\gamma\} \circ w(A) \subseteq w(B)\}$.

Here for any two sets $\mathcal{A} \subseteq \mathcal{W}^+$ and $\mathcal{B} \subseteq \mathcal{W}^+$ by $\mathcal{A} \circ \mathcal{B}$ we denote the set $\{\alpha \circ \beta \mid \alpha \in \mathcal{A} \text{ and } \beta \in \mathcal{B}\}$.

Definition. A sequent $A_1 \ldots A_n \to B$ is *true* in a model $\langle \mathcal{W}^+, \circ, w \rangle$ iff

$$w(A_1) \circ \ldots \circ w(A_n) \subseteq w(B).$$

Theorem 2. *A sequent is derivable in the Lambek calculus if and only if it is true in every L-model.*

Theorem 3. *A sequent is derivable in the Lambek calculus if and only if it is true in every L-model over an alphabet $\mathcal{W}$ consisting of two symbols.*

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